PassFrame tuition

Factors, Multiples & Primes

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 1, 2, 4, 8

2. 1, 3, 5, 15

3. 1, 2, 4, 5, 10, 20

4. 3, 6, 9, 12, 15

5. 7, 14, 21, 28, 35

6. Yes

7. No (1, 3, 7, 21)

8. 2, 3, 5, 7

9. 45

10. 1, 2, 3, 6, 9, 18

11. Yes

12. 11, 22, 33, 44

13. 1, 2, 3, 4, 6, 8, 12, 24

14. 18

15. 5

Standard

1. 12

2. 14

3. 36

4. 42

5. $2 \times 2 \times 3 \times 7$

6. $2 \times 3 \times 3 \times 5$

7. 1, 2, 3, 4, 6, 12

8. 24

9. No ($119 = 7 \times 17$)

10. 53

11. 12

12. 40

13. (1,60) (2,30) (3,20) (4,15) (5,12) (6,10)

14. Yes (1+0+8=9; $108 = 9 \times 12$)

15. 36

Standard / Hard

1. 36 seconds

2. 12 bags

3. 30

4. 48

5. 41

6. 2 (it is even and prime)

7. 15

8. 24

9. 12 m × 12 m squares

10. 97

11. 10:00am (60 minutes later)

12. $2 \times 2 \times 3 \times 5$

13. 11 and 13 (one valid pair)

14. 18

15. 4 (2^2 — factors 1, 2, 4)

Homework — Answers

Easy / Standard

1. 1, 2, 5, 10

2. 1, 2, 7, 14

3. 1, 5, 25

4. 4, 8, 12, 16, 20

5. 9, 18, 27, 36, 45

6. Yes

7. No (3×9)

8. 11, 13, 17, 19

9. 56

10. 1, 2, 4, 8, 16

11. No (3×3)

12. 12, 24, 36, 48

13. 1, 2, 4, 7, 14, 28

14. 24

15. 8

Standard

1. 8

2. 15

3. 72

4. 30

5. 2 × 2 × 2 × 3 × 3

6. 2 × 2 × 5 × 5

7. 1, 2, 4, 5, 10, 20

8. 36

9. No (133 = 7 × 19)

10. 61

11. 24

12. 30

13. (1,72) (2,36) (3,24) (4,18) (6,12) (8,9)

14. Yes (1+2+6=9; 126 = 9 × 14)

15. 30

Standard / Advanced

1. 6 (LCM(4,6)=12, HCF(12,30)=6)

2. 1, 11 (11 is the smallest 2-digit prime)

3. 30 (2×3×5)

4. Yes, 7 (HCF(35,49)=7)

5. 6

6. 24 minutes

7. 18 bouquets

8. 24 pupils

9. 12 cm

10. 73

11. 30 days

12. 108

13. 14 m

14. 42 boxes (2 croissants and 3 pastries per box)

15. 5 times (0, 72, 144, 216, 288 seconds)